

# STEPHANIE R. KILLINGSWORTH

PhD Candidate – University of Florida



[skillingsworth@floridamuseum.ufl.edu](mailto:skillingsworth@floridamuseum.ufl.edu)



<https://orcid.org/0000-0001-5257-986>



<https://github.com/stephkillingsworth>



[www.stephaniekillingsworth.com](http://www.stephaniekillingsworth.com)

---

## EDUCATION

### **2024-Present | PhD in Geology - Paleontology | University of Florida**

*Dissertation in progress (expected completion 2027 December): Macroevolutionary Transitions in Neogene Fossil Horses (Family Equidae) in North America Using Phenomics and Machine Learning*

Advisors: Drs. Bruce J. MacFadden and Arthur Porto

### **2023 | MS in Geology - Paleontology (Summa cum laude) | University of Florida**

*Thesis: Geochemistry of Fossil Shark Teeth as Age Proxies for Neogene Sites of Florida*

Advisor: Dr. Bruce J. MacFadden

### **1997 | Graduate Coursework in Microbiology (Degree Not Awarded) | University of Kansas**

### **1995 | BA in Human Biology (Cum laude) | University of Kansas**

---

## PROFESSIONAL EXPERIENCE

### **2019-Present | K-12 Education and Outreach Coordinator**

University of Florida, Thompson Earth Systems Institute - Florida Museum of Natural History

### **2013-2019 | Certified 6-12 Biology Teacher State of Florida**

**Science Department Chair and Science Night Coordinator**

Palm Beach County School District

### **2001-2013 | Orthopedic/Neurologic Clinical Specialist and Sales**

Dynasplint Systems Inc. and Depuy Johnson and Johnson

---

## PEER REVIEWED PUBLICATIONS

\* Denotes scientific research    \*\* Denotes education research and Broader Impacts

## 2026

1. Ivory III, A. A., Ville-Veikko, Killingsworth, S., & Donovan, V. M. Integrating Scientific Tools into K-12 Education: Activities, Technologies, and Partnerships. *EDIS*, Gainesville, Florida. In Press. \*\*
2. Wusylko, C., Still, R., Antonenko, P. D., Abramowitz, B., Perez, V., Waisome, J., **Killingsworth, S.**, & MacFadden, B. Exploring Bias and Building AI Models: Data Science for Middle Schoolers. *Science Scope*. <https://doi.org/10.1080/08872376.2026.2632617> \*\*

## 2025

3. MacFadden, B. J., Barberis, C., Vallejo-Pareja, M. C., Zbinden, S. P., Perez, V. J., **Killingsworth, S. R.**, Marks, K., Hall, D., & Porto, A., 2025. AI and paleontology: Effects of vertebrate fossils sample size on machine learning image classification. \*
4. **Killingsworth, S.**, MacFadden, B., Moran, S. M., Perez, V. J., Pirlo, J., & Ziegler, M. Marine strontium isotopes preserved in fossil shark teeth calibrate Neogene land mammal evolution. *Palaeogeography, Palaeoclimatology, Palaeoecology*. <https://doi.org/10.1016/j.palaeo.2024.112698> \*
5. Wusylko, C., Antonenko, P., Abramowitz, B., Waisome, J., Perez, V., **Killingsworth, S.**, & MacFadden, B. Supporting teachers to integrate AI and computer science through SharkAI: a middle school computer science and paleontology curriculum. *Journal of Technology and Teacher Education*. <https://doi.org/10.70725/161207qresyx> \*\*



## 2024

6. MacFadden, B. J., Bohaska, D., Cone, L., **Killingsworth, S. R.**, Zbinden, S. P., Moran, S. M., & Perez, V. J. Early Miocene land mammals and chronology of the Belgrade Formation, eastern North Carolina. *Journal of Paleontology*. <https://doi.org/10.1017/jpa.2024.68> \*
7. Abramowitz, B., Ennes, M., Antonenko, P., & **Killingsworth, S.** A Narrative Inquiry into Teacher Efficacy for Teaching Climate Science with Technology in a Scientist-Teacher Partnership Program. *The Journal of Environmental Education*. <https://doi.org/10.1007/s10956-024-10169-x> \*\*



## 2023

8. **Killingsworth, S.**, MacFadden, B. Species Occurrences of Late Miocene Horses (Equidae) from Florida: Sampling, Ecology, or Both? *Paleobiology*. <https://doi.org/10.1017/pab.2023.35> \*



## 2021

9. Abramowitz, B., Ennes, M., **Killingsworth, S.**, Antonenko, P., MacFadden, B., & Ivory, A. Science in School: Transforming K-12 Outreach through Scientist Teacher Partnerships. *Journal of STEM Outreach*. <https://doi.org/10.15695/jstem/v4i1.14> \*\*



## 2020

10. MacFadden, B. J., Pirlo, J., Abramowitz, B., **Killingsworth, S.**, & Ziegler, M. Authentic field experiences for STEM teachers: Collecting Florida fossils. *Florida Association of Science Teachers*. <https://doi.org/10.1101/2020.07.02.184077> \*\*




---

## POPULAR ARTICLES

1. Killingsworth, S., MacFadden, B., 2025. Fossil shark teeth are abundant and can date the past in a unique way. *The Conversation*. <https://doi.org/10.64628/aai.sysf9ypn6>
2. **Killingsworth, S.**, MacFadden, B., 2024. Horses lived in the Americas for millions of years – new research helps paleontologists understand the fossils we’ve found and those that are missing from the record. *The Conversation*. <https://doi.org/10.64628/aai.w4vhe6cwp>




---

## FIELD AND TECHNICAL EXPERIENCE

### *Field Collecting*

- |              |                  |              |
|--------------|------------------|--------------|
| ● Florida    | ● Nebraska       | ● California |
| ● New Mexico | ● North Carolina | ● Panama     |

### *Museum Collections Vertebrate Paleontology*

- |                                     |                                      |                                                |
|-------------------------------------|--------------------------------------|------------------------------------------------|
| ● Florida Museum of Natural History | ● American Museum of Natural History | ● Natural History Museum of Los Angeles County |
|-------------------------------------|--------------------------------------|------------------------------------------------|

- Panhandle Plains Historic Museum
- Smithsonian National Museum of Natural History
- Berkeley Museum of Natural History

### **Technical Computer Experience**

- Microsoft Suite, Google Suite, Video Conferencing Platforms, Sharepoint, Canvas, Qualtrics, Adobe Illustrator, WordPress, MailChimp, Slack, Discord, SciWheel
- R/RStudio, Python, Rev, HTML, GitHub, RevBayes, ArcGISPro, 3DSlicer, PBDB, GBIF, iDigBio, Specify, HiperGator Supercomputer Cluster

### **Technical Lab Experience**

- Hood and bench practice and safety, analytical balances, media preparation, aseptic techniques, tissue culture, centrifuge, autoclave, incubation, microscopy, microtome, PCR, gel electrophoresis
- Column chemistry, ICP-MS, REEs, Isotope analysis (Sr), micro-CT

---

## **CONFERENCES & SPEAKING ENGAGEMENTS**

1. **Killingsworth, S.**, 2026. From Pixels to Morphospace: Interpretable Deep Learning Reveals Paleopopulation Structure and Evolutionary Variation in Neogene Horses (Family Equidae). SVP, Cleveland, Ohio. *Romer Prize Session*
2. Broo, J., & Neilson, A., Wusylko, C., & **Killingsworth, S.**, 2026. Using machine learning models to assess variation in fossilized horse teeth for species classification in education. SVP, Cleveland, Ohio.
3. **Killingsworth, S.**, 2026. Museums as Resources for Rural Education. Southwest Regional Community Engagement Consortium Convening. *Invited Panelist* El Paso, Texas.
4. **Killingsworth, S.**, 2025. Detecting Macroevolutionary Transitions in Neogene Fossil Horses. *Invited Lightning Talk* at the Florida Museum of Natural History.
5. **Killingsworth, S.**, 2025. A Trail Ride Through Time: Epic Adventures and Neogene Horse Evolution. *Invited Talk* at University of California, Berkeley.
6. Haire, G., Antonenko, P., Wusylko, C., **Killingsworth, S.**, Abramowitz, B., Waisome, J., Jones, D., MacFadden, B., & Perez, V., 2025. Exploring Student Self-Efficacy in AI Through Model Building Artifacts. American Society of Engineering Education.
7. Wusylko, C., Still, R., Antonenko, P., Abramowitz, B., Perez, V., Waisome, J., **Killingsworth, S.**, & MacFadden, B., 2025. One Teacher's Experience Adapting and Innovative, Flexible Computer Vision Curriculum in a Science Classroom. American Society of Engineering Education.
8. Opoku, R., Jones, T., Haire, G., Wusylko, C., Lin, C. H., Adhami, N., MacFadden, B., Waisome, J., Perez, V., Abramowitz, B., **Killingsworth, S.**, King, N., & Antonenko, P.,

2025. Black Middle Schoolers' Artificial Intelligence Self-Efficacy and Outcome Expectations in a Summer STEM Camp. AERA Annual Meeting. Denver, Colorado.
9. Jones, T., Opoku, R., Haire, G., Wusylko, C., Lin, C. H., Adhami, N., MacFadden, B., Waisome, J., Perez, V., Abramowitz, B., **Killingsworth, S.**, & Antonenko, P., 2025. Exploring Black Girls' Sources of Self-Efficacy for Artificial Intelligence During an Informal Learning Experience. AERA Annual Meeting. Denver, Colorado.
  10. Haire, G., Jones, T., Opoku, R., Wusylko, C., Lin, C. H., Adhami, N., MacFadden, B., Waisome, J., Perez, V., Abramowitz, B., **Killingsworth, S.**, & Antonenko, P., 2025. Secondary Students' Outcome Expectations for AI: A Social Cognitive Career Theory Perspective. AERA Annual Meeting. Denver, Colorado.
  11. **Killingsworth, S.**, 2024. Harnessing Machine Learning to Detect Macroevolutionary Transitions in Neogene Fossil Horses (Family Equidae) in North America. SVP Conference. Minneapolis, Minnesota.
  12. Perez, V., MacFadden, B. J., Antonenko, P., Waisome, J., Abramowitz, B., **Killingsworth, S.**, & Vallejo-Pareja, M., 2024. The Shark AI Project: Utilizing Paleontology to Catalyze Learning About Artificial Intelligence. GSA Conference. Anaheim, California.
  13. Wusylko, C., Antonenko, P., Abramowitz, B., Waisome, J., Perez, V., **Killingsworth, S.**, & MacFadden, B., 2024. Supporting Teachers to Integrate AI and Computer Science Through SharkAI: A Middle School Computer Science and Paleontology Curriculum. SITE Interactive. Online.
  14. **Killingsworth, S.**, 2023. Geochemistry of Fossil Shark Teeth as Age Proxies for Neogene Sites of Florida. GSA Conference. Pittsburgh, Pennsylvania
  15. **Killingsworth, S.**, & MacFadden B., 2023. Species Occurrences of Late Miocene Horses (Equidae) from Florida: Sampling, Ecology, or Both? SVP Conference. Cincinnati, Ohio.
  16. MacFadden, B., Antonenko, P., Perez, V., Abramowitz, B., **Killingsworth, S.**, & Parnell, D., 2023. Fossil Shark Teeth Activate Engagement and Learning About AI (Artificial Intelligence) in Florida Public Middle Schools. SVP Conference. Cincinnati, Ohio.
  17. Sharma, D., **Killingsworth, S.**, Bokor J., Pirlo J., Abramowitz, B. & MacFadden, B., 2023. Chewing on Change: Training the Next Generation of Science Communicators. SVP Conference. Cincinnati, Ohio.
  18. **Killingsworth, S.**, Abramowitz, B., MacFadden B., Carter, K., & Pirlo, J., 2022. Bringing Florida Museum of Natural History 3D Collections into K-12 Schools to Teach About Adaptations, Evolution, and Climate Change. SPNHC Conference. Edinburgh, Scotland.
  19. MacFadden, B., Antonenko, P., Magruder, J., Perez, V., Abramowitz, B., & **Killingsworth, S.**, 2022. Integrating AI Machine Learning into the Teaching of Paleontology Using Fossil Shark Teeth in Middle Schools. Poster presentation at AI in Education Mini-Symposium. Gainesville, Florida.

20. Abramowitz, B., & **Killingsworth, S.**, 2021. Implementing Machine Learning and AI to Advance Science Curriculum. Presentation at Florida Association of Science Teachers. St. Augustine, Florida.
21. Ivory, A., Abramowitz, B., & **Killingsworth, S.**, 2021. The Water that Connects Us. Presentation at Florida Association of Science Teachers. St. Augustine, Florida.
22. Abramowitz, B., Ennes, M., **Killingsworth, S.**, Antonenko, P., MacFadden, B., & Ivory, A., 2021. Engaging scientists in outreach to support Florida teachers. Presentation at the Southeast Association of Science Teacher Education. Tampa, Florida.
23. MacFadden, B. & **Killingsworth, S.**, 2021. Program Updates From the Scientist in Every Florida School at Florida Association of Science Supervisors. Miami, Florida.
24. Abramowitz, B., MacFadden, B. J., Pirlo, J., **Killingsworth, S.**, & Ziegler, M., 2021. Structuring and Authentic Teacher Professional Development Experience. Poster presented at UF College of Education Research Symposium, Gainesville, Florida.
25. **Killingsworth, S.**, Abramowitz, B., Valle-Levinson, A., 2020. Exploring air, water, land and life with Scientist in every Florida school. Virtual presentation at Florida Association of Science Teachers. Virtual
26. MacFadden, B., Abramowitz, B., & **Killingsworth, S.**, 2020. TESI and Scientist in Every Florida School. Invited Presentation for the Fall 2020 Geology Seminar Series. University of Florida. Gainesville, FL.
27. **Killingsworth, S.**, & Abramowitz, B., 2019. K-12 Outreach, Moonshot Project. Invited Presentation for the Fall 2020 Broader Impacts of Science on Society graduate course. University of Florida. Gainesville, Florida.
28. MacFadden, B. & **Killingsworth, S.**, 2018. Scientist in Every Florida School and Teacher Experiences at Florida Association of Science Supervisors.
29. **Killingsworth, S.** & Bokor, J., 2016. Summer Science Institute at CPET in Gainesville at the Palm Beach County STEM Initiative Learning Showcase. West Palm Beach, Florida.
30. **Killingsworth, S.**, 2016. Experimental Science and Engineering and Design Lab/Project at the Creativity Fest. West Palm Beach, Florida.
31. **Killingsworth, S.**, 2015. Translating the GABI-RET Experience into the Middle School Classroom at the 2015 PCP PIRE/GABI RET/FOSSIL Collaborative Field Conference. Gainesville, Florida.

---

## GRANT COLLABORATIONS

\$5,000

NAS Inspire Award | ECR Fellows to Connect with K-12 Students in the Gulf Region | 2025

|           |                                                                                                                                                                                                                                                                                                                                      |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$12,959  | Bureau of Land Management Subaward   Your Public Lands at Risk- Archaeological Exploration of the Jupiter Inlet and Lighthouse   2025                                                                                                                                                                                                |
| \$21,000  | Dickinson Fund   Florida Museum of Natural History Vertebrate Paleontology   2024                                                                                                                                                                                                                                                    |
| \$48,128  | Senior Personnel Subaward   NSF CAREER #AWD16116: An Integrated Trustworthy AI Research and Education Framework for Modeling Human Behavior in Climate Disasters   Grant Total: \$501,181   P.I. Zhao, X.   UF Engineering School of Sustainable Infrastructure and Environment   2024                                               |
| \$118,170 | Senior Personnel Subaward   NSF#DBI-2320251: BII: Polyploidy: Integration Across Scales and Biological Systems   Grant Total: \$12,667,073   P.I. Soltis, P. & others   UF Department of Natural History   2023                                                                                                                      |
| \$13,214  | Senior Personnel Subaward   NSF#AWD12799: Research Infrastructure: Moving and Improving the Florida Museum Fish Collection   Grant Total: 695,318   P.I. Page, L., UF Department of Natural History-Ichthyology   2022                                                                                                               |
| \$152,830 | Senior Personnel Subaward   NSF #2147625: Innovative Technology Experiences for Students and Teachers (ITEST). DTI: Teaching Middle School Students Machine Learning Concepts and Applications in Science with Fossil Shark Teeth   Grant Total: \$1,275,109   P.I. MacFadden, B. & Others   UF Department of Natural History   2022 |
| \$5,500   | Paleontological Society and Alachua School District   Chewing on Change: Bringing Fossil Horses into Title I Schools to Learn About Macroevolution   2022                                                                                                                                                                            |
| \$80,132  | Senior Personnel Subaward   NSF#2114833: Intramembrane Proteolysis in Regulation of the Rice Receptor Kinase XA21   Grant Total: \$945,996   P.I. Song, W. Y.   2021                                                                                                                                                                 |
| \$15,000  | Surface 71   Palm Beach Philanthropy Tank   2018                                                                                                                                                                                                                                                                                     |

---

## AWARDS AND NOMINATIONS

### *National Awards*

\$300 – University of Florida Division III Superior Achievement Award (2026)  
\$2000 – Winifred Goldring Award- Assoc. for Woman Geoscientists & Paleo. Society (2023)  
NSF Presidential Awards for Excellence in Mathematics & Science Teaching Finalist (2019)  
\$10,000 – National Life Group Life Changer of the Year (2017-2018)

### ***Research and Travel Awards***

\$400 – University of Florida CLAS Travel Grant (2026)  
\$1000 – Department of Geological Sciences (UF) Research (2025)  
\$1750 – Theodore Roosevelt Memorial Fund (AMNH) Grant (2025)  
\$500 – Department of Geological Sciences (UF) Research (2024)  
\$1000 – Florida Museum of Natural History Spring Travel Award (2023)  
\$500 – Jackson School of Geosciences Student Member Travel Grant (SVP) (2023)  
\$350 – Department of Geological Sciences (UF) Travel to GSA (2023)  
\$500 – Graig D. Shaak Staff Enrichment Award- Florida Museum of Natural History (2023)  
\$300 – Nichol Award- University of Florida Department of Geological Sciences (2023)  
\$1000 – Mitchell Hope Award- Southwest Florida Fossil Society (2023)  
\$500 – Travel Award for Society for the Preservation of Natural History Collections (2022)

### ***Teaching Awards***

\$6000 – Florida's Best and Brightest Recipient-Highly Effective Rating (2016-2019)  
\$1500 – Great American Biotic Interchange Research Experience for Teachers (2014-2019)  
\$1000 – Dwyer Teacher of the Year Runner Up (2018)

### ***Distinctions***

Romer Prize Session SVP (2026)  
Western Kentucky National STEM Scholar (2019)  
Dwyer Teacher of the Year (2018, 2019)  
National Geographic Grosvenor Fellow Nominee (2018, 2019)  
Macy's Teacher of the Year (2016)

---

## **LEADERSHIP**

### ***PROFESSIONAL DEVELOPMENT K-12 PROGRAM INITIATIVES***

**Killingsworth, S.,** Zhao, X., 2026-2027. *Modeling Human Behavior in Climate Disasters Using Artificial Intelligence*

**Killingsworth, S.,** Musgrave, P., 2026-2027. *Experimental Design Using Robotic Fish*

**Killingsworth, S.,** Neilson, A., 2026-2027. *Archaeological Exploration of the Jupiter Inlet*

**Killingsworth, S.,** Neilsen, A., & Wuslyko, C., 2026. *Chewing on Change - Using Machine Learning Models to Assess Variation in Fossilized Horse Teeth For Species Classification*

**Killingsworth, S.,** Xue, B., & Leibold, M., 2025. *Ecological Competition Between Invasive and Native Species*

**Killingsworth, S.,** Soltis, P., & Soltis, D., 2025-2028. *BII: Polyploidy: Integration Across Scales and Biological Systems*

**Killingsworth, S.,** & Ennes, M., 2023-2024. *Research Infrastructure: Moving and Improving the Florida Museum Fish Collection*

**Killingsworth, S.,** Abramowitz B., Bokor, J., Pirlo, J., & MacFadden, B., 2022. *Chewing on Change: Bringing Fossil Horses into Title I Schools to Learn About Macroevolution*

**Killingsworth, S.** & Abramowitz B., MacFadden, B., Antonenko, P, Waisome, J., 2022-2025. *SEFS Cohort I-IV: AI Learning in K-12 with Fossil Sharks*

**Killingsworth, S.,** Song, W. Y., 2022-2025. *SEFS Cohorts I-IV: Plant Pathology Professional Development for High School Teachers*

**Killingsworth, S.,** Abramowitz B., Pirlo, J., & MacFadden, B., 2022. *SEFS Paleo PD*

**Killingsworth, S.,** Abramowitz B., Carter K., & MacFadden, B., 2022. *Adaptations, Evolution, and Climate Change Miami-Dade Teacher Professional Development*

**Killingsworth, S.** & Abramowitz B., 2021. *The Hydrosphere Summer Professional Development for Teachers*

**Killingsworth, S.,** Abramowitz B., Pirlo, J., & MacFadden, B., 2021. *SEFS Paleo PD*

**Killingsworth, S.** & Abramowitz B., 2020. *The Nature of Science Summer Professional Development for Teachers*

**Killingsworth, S.,** Abramowitz B., & MacFadden, B., 2020. *World Wetlands January Summit Teacher Professional Development*

**Killingsworth, S.** 2019. Florida Association of Science Teachers Conference. *The Case of Coral*

**Killingsworth, S.,** Abramowitz B., Pirlo, J., & MacFadden, B., 2019. *Connections to the Biosphere Summer Professional Development*

**Killingsworth, S.,** Abramowitz B., Pirlo, J., & MacFadden, B., 2019. *SEFS Montbrook Dig Experience Professional Development*

---

## **OUTREACH & SERVICE**

2025-Present                      Non-Profit Big Dog Ranch Rescue volunteer

2019-Present                      Scientist Outreach - Scientist in Every Florida School

**MUSEUM MENTORSHIP/LEADERSHIP**

|      |                                                                   |
|------|-------------------------------------------------------------------|
| 2025 | Mentoring/Supervising undergraduate Alexa Neilson                 |
| 2023 | Mentoring undergraduate student: Sam Zbinden                      |
| 2023 | Selection Committee Member for VP Assistant Curator for the FLMNH |

**EDUCATION MENTORSHIP/LEADERSHIP**

|           |                                                                      |
|-----------|----------------------------------------------------------------------|
| 2020-2022 | Presidential Award of Excellence in Math & Science Teaching Judge    |
| 2019-2022 | Florida Association of Science Teachers (FAST) Area 10 Director      |
| 2019-2022 | National Association of Geoscience Teachers (NAGT) SE Representative |

**JOURNAL REVIEWS**

|      |                                                    |
|------|----------------------------------------------------|
| 2026 | Manuscript Reviewer <i>Springer PalZ journal</i>   |
| 2025 | Manuscript Reviewer <i>Paleobiology</i>            |
| 2023 | Manuscript Reviewer <i>Journal of Paleontology</i> |

---

**PROFESSIONAL MEMBERSHIP**

Paleontological Society  
Society for Vertebrate Paleontology  
American Women in Geosciences (Northern Florida Chapter)  
Geological Society of America  
American Alliance of Museums